Vijay Magadum

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Summary

Embedded Systems Engineer with hands-on experience in bare-metal and RTOS-based firmware development on ESP32 and ARM7 platforms. Proficient in industrial communication protocols including Modbus, CAN, MQTT, and UART, with practical experience in PCB bring-up, hardware debugging, and IoT product development. Delivered multiple client-deployed embedded systems in industrial automation and IIoT domains within the first year of professional experience. Seeking a product-focused embedded role to deepen expertise in firmware architecture and connected systems.

Technical Skills

Languages	C, C++, Embedded C
Microcontrollers	LPC2129 (ARM7), ESP32
Protocols	MODBUS, CAN, I^2C, SPI, UART, MQTT, TCP/IP
RTOS & OS	FreeRTOS, Linux Internals
Tools	Keil µVision5, Arduino IDE, Proteus, Flash Magic, Git, VS Code
Hardware	PCB Assembly & Bring-up, Debugging.

Work Experience

Embedded Engineer Trainee — EmbedSol Technologies LLP Jun 2025 – Present

- Developed and delivered client-facing firmware in C/C++ for ESP32-based industrial automation and IoT products, ensuring reliable real-time operation across multiple hardware configurations.
- Validated hardware and firmware functionality through structured test plans and register-level debugging, resolving critical defects before client deployment.
- Configured and integrated IoT gateways for client deployments, enabling seamless MQTT-based device communication and remote data access.

Embedded Systems Intern — Vector India, Bengaluru Aug 2024 – May 2025

- Built bare-metal embedded applications on ARM7 (LPC2129), implementing low-level drivers for GPIO, timers, PWM, and peripheral interfacing from scratch.
- Implemented and validated CAN, I^2C, SPI, and UART protocols in real-time systems; debugged protocol behavior using oscilloscopes and logic analyzers.
- Explored Linux system programming internals — processes, signals, IPC, memory management, and scheduling.

Projects

Industrial Water Monitoring & Pump Control System *ESP32, MODBUS, MQTT, GSM*

Multi-tank water parameter monitoring and remote pump control system. Developed firmware for sensor acquisition via MODBUS, real-time data publishing over MQTT through GSM, and relay-based pump control logic.

Manufacturing Safety Interlock Controller *ESP32, Flash Storage, PLC*

Fail-safe production line interlock that halts machine operation until defective parts are confirmed removed, with persistent fault logging and PLC handshake signaling.

Industrial IoT Temperature Logger *ESP32, MQTT*

Temperature monitoring system with MQTT telemetry, local data logging, OTA firmware updates, and fault-tolerant architecture

Education

Apr 2024 Bachelor of Engineering, Electrical Engineering

Dr. D.Y. Patil Pratishthan's College of Engineering, Kolhapur **CGPA: 7.76**